

MICHELLE GUROVITH
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SUMMARY

Fullstack software engineer with an M.S. in Computer Science and Engineering from UC Santa Cruz. Experienced building production APIs, real-time data systems, and distributed architectures. Proficient in Python, C++, React, and TypeScript, with hands-on experience in SQL and NoSQL databases.

EDUCATION

M.S. Computer Science and Engineering: University of California, Santa Cruz

- September 2025 – June 2026

B.S. Computer Engineering: University of California, Santa Cruz

- September 2022 – June 2025

EXPERIENCE

Parlo – Software Engineer: September 2025 – February 2026

- Engineer a real-time synchronization layer in Firebase/Firestore supporting concurrent interactions across 150+ active users.
- Redesign Firestore data models and batch write workflows to eliminate race conditions and ensure consistent application state across clients.
- Optimize database query patterns, indexing strategies, and read/write operations, contributing to a 17% increase in daily active users.
- Design scalable backend architectures that support rapid feature development while maintaining reliability and performance.
- Build internal tooling and monitoring solutions to identify bottlenecks, improve system stability, and accelerate debugging.

BiometerPlus – Software Engineer Intern: June 2024 – September 2024

- Built token-based authentication and account management systems backed by MySQL, validated through comprehensive Postman test suites prior to release.
- Designed and implemented RESTful API endpoints with MySQL integrations supporting biometric data collection, storage, and retrieval.
- Developed a secure password recovery workflow with time-limited tokens and server-side validation, preventing unauthorized access through expired or replayed credentials.
- Implemented full account lifecycle management, including soft deletion, reactivation, and JSON schema validation, with RFC-compliant HTTP status handling across all endpoints.
- Enhanced backend reliability and data integrity through input validation, authentication controls, and structured error-handling mechanisms.

RESEARCH AND PROJECTS

JazzyFS: Adaptive Experimental Filesystem (Master's Thesis)

- Designed and implemented a FUSE-based filesystem with adaptive file prefetching and runtime access analysis
- Built instrumentation and benchmarking tools to evaluate filesystem performance across multiple workload types
- Developed a sonification system that converts filesystem activity into real-time audio feedback and visualizations
- Created automated experiment pipelines for repeatable testing, performance analysis, and result generation

Fault-Tolerant Distributed Sensor Network

- Developed a fault-tolerant token-ring communication system across three Raspberry Pi devices
- Built a MySQL-based data storage system for distributed sensor data collection and management
- Created automated monitoring and alerting tools to detect inactive devices and improve system reliability
- Implemented communication and data transfer systems for reliable cross-device synchronization

TECHNICAL SKILLS

- **Languages:** Python, C++, TypeScript, SQL
- **Frameworks and Libraries:** React, FUSE
- **Databases:** PostgreSQL, MySQL, Firebase/Firestore
- **Tools and Infrastructure:** Linux, Docker, Git, GitHub Actions, Bash, Postman

ADDITIONAL

- Advanced Certificate of Merit in Classical Piano
- Composed and copyrighted original music